D115-D116 | Denominators, costs and uncertainty

PRIVATE FICTIONAL cases. No actual fraud/customer labels, experiment, user data, effect evidence or deployment authority. Attempt independently before reference.

D115 | Rare-event denominators	In a synthetic labelled 10000 batch, 100 fraud and 9900 legitimate, sensitivity 90% and specificity 95% give TP 90 / FN 10 / FP 495 / TN 9405. PPV 90/585=15.384615%, not sensitivity 90%. Always-negative accuracy 99% still misses 100 frauds.
D115 | Review policy is stipulated	Perfect review of every flag before loss costs 20 per flag; legitimate flag friction 30; missed loss 1000. Policy 36550 versus no-screen 100000, modeled reduction 63450 before fixed/privacy/appeals costs. Break-even missed loss 295; loss 200 makes policy worse 8550. Fractional changed-case cells are expected counts, not observed labels.
D116 | Fixed-horizon inference	Independent users, one binary outcome per user, prespecified window/metric, no peeking or variants. A 100/1000 and B 102/1000: absolute 0.002=0.2 percentage points and relative 2%. Unpooled Wald CI about[-2.441248,2.841248] percentage points. Pooled-null two-sided p about 0.8820 is not probability null true.
D116 | Prospective planning	Baseline 10% with 2% relative target gives 10.2%, not 12%. Specified approximation z 1.96+0.84: 355933 complete per arm, 711866 total. Two-percentage-point target instead 3838 per arm. At 10% stipulated noninformative attrition, 395482 randomized per arm. This formula is not exact binomial power.
Scope and stop	No observed learner attempts/errors/reviewer words or delayed transfer. Static prose is Case-v1 and needs revision after edits. Authored LibreOffice execution only; no Excel/Sheets parity, new empirical source reading, full AV, learner/account/legal clearance or overall 10/10. No real screening, experiment, data collection, contact, disclosure, blocking or spend authorized.

D115 inputs | Define the deployment denominator

Batch size integer; prevalence/sensitivity/specificity are stipulated fractions. Review assumed perfect before loss. Costs INR per transaction/event, not annual cash value.

Synthetic batch size	10000.00
Fraud prevalence	0.01
Sensitivity: P(flag | fraud)	0.90
Specificity: P(no flag | legitimate)	0.95
Perfect review cost per flag, INR	20.00
Additional friction per legitimate flag, INR	30.00
Loss per missed fraud, INR	1000.00

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D115 matrix | Conditioning changes the denominator

Expected counts under stipulated rates. Baseline integer counts coincide with the original labelled toy batch. Odds/LR route may be undefined at boundary rates even when direct PPV is defined.

Measure	Value
Fraud denominator	100.000000
Legitimate denominator	9900.000000
True positives, TP	90.000000
False negatives, FN	10.000000
False positives, FP	495.000000
True negatives, TN	9405.000000
Flagged denominator	585.000000
Not-flagged denominator	9415.000000
PPV: fraud / flags	15.3846%
False discovery: legitimate / flags	84.6154%
NPV: legitimate / not flagged	99.8938%
Accuracy: correct / all	94.9500%
Always-negative accuracy	99.0000%
Prior fraud odds	0.010101
Positive likelihood ratio	18.000000
Posterior odds, via odds times LR	0.181818
Bayes probability, via posterior odds	15.3846%

A balanced case-control sample cannot supply target prevalence unaided. Transported rates and perfect labels are assumptions, not measured generalization. Zero flags makes PPV/false discovery undefined; zero nonflags makes NPV undefined. Specificity 1 makes LR denominator zero, not an error to conceal.

D115 policy | Accuracy does not price consequences

Perfect review catches all true positive fraud before loss, no additional true-positive cost or benefit. FN loss only. No screening/review/privacy policy grant.

Measure	Value
Review cost, INR	11700.000000
Legitimate flag friction, INR	14850.000000
Missed-fraud loss, INR	10000.000000
Total policy cost, INR	36550.000000
No-screen missed loss, INR	100000.000000
Modeled cost reduction, INR	63450.000000
Break-even loss, INR per fraud caught	295.000000
Cost accounting identity residual	0.000000

Baseline policy 36550 versus no-screen 100000. Loss threshold 295 assumes fixed flags and friction. At loss 200, policy 28550 versus 20000: worse 8550. No detected fraud makes finite break-even undefined. Fixed setup, delay, appeals, privacy and imperfect review omitted, not free.

D116 inputs | One fixed-horizon synthetic batch

Independent users, one binary outcome each; same window, no peeking/variants/attrition in this completed batch. Counts alone cannot prove assignment, independence or causal validity.

Arm A analyzed users, integer	1000.00
Arm A conversions, integer	100.00
Arm B analyzed users, integer	1000.00
Arm B conversions, integer	102.00

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D116 inference | Imprecision is not equivalence

Unpooled Wald interval versus pooled-null z test. Normal approximation not exact. Count screen is not proof of randomization, independence, valid stopping, no bias or sufficient precision.

Measure	Value
Observed conversion A	10.0000%
Observed conversion B	10.2000%
Absolute difference B minus A	0.2000%
Relative lift, if A rate positive	2.0000%
10 success/failure count screen (1=pass)	1.000000
Unpooled CI standard error	0.013476
Wald 95% lower, nominal z=1.96	-2.4412%
Wald 95% upper, nominal z=1.96	2.8412%
Pooled null proportion	10.1000%
Pooled null standard error	0.013476
Pooled null z	0.148414
Two-sided normal-approximation p	0.882016
Observed interval width, proportion units	0.052825
Observed difference, percentage points	0.200000

CI contains zero, harmful effects and useful positive effects. Nonsignificance is not equivalence; that needs prespecified margins/design/test. A realized frequentist interval is not 95% posterior parameter probability. No retrospective observed power or probability null true. Extreme z may numerically round p to 0; not exact zero.

D116 inputs | Prospective targets, not observed power

Specified approximate equal-allocation normal planning formula, not exact binomial power. z 1.96 and 0.84 are course constants for nominal two-sided 5%/80% planning.

Prospective baseline conversion	0.10
Relative target lift, signed fraction	0.02
Separate absolute target, proportion units	0.02
Specified nominal alpha/2 z	1.96
Specified target power z	0.84
Independent noninformative attrition fraction	0.10

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D116 planning | Per arm is not total

Relative target and separate percentage-point target use distinct alternatives/pbar. Zero delta makes required n undefined. Negative nonzero targets use squared delta; no direction/power guarantee.

Measure	Relative target	Absolute target
Prospective alternative rate	10.2000%	12.0000%
Signed absolute target delta	0.2000%	2.0000%
Average planning rate	10.1000%	11.0000%
Approximate required complete n PER ARM	355932.080000	3837.680000
Ceiling complete users PER ARM	355933.000000	3838.000000
Total complete users, both arms	711866.000000	7676.000000
Ceiling randomized PER ARM after attrition	395482.000000	4265.000000
Total randomized users	790964.000000	8530.000000
Approximate raw n ratio: relative / absolute	92.746680	92.746680

Baseline 355932.08 -> 355933 complete PER ARM,711866 total; after 10% attrition 395482 per arm. Absolute target 3837.68 -> 3838 per arm. Raw n ratio 92.74668, not 100 because pbar changes. Attrition inflation cannot fix missing-outcome bias. High N cannot fix faulty assignment or measurement.

Audit | Three independent arithmetic cases

Flags test internal identities only. They do not verify real labels, costs, representative data, valid experimentation, statistical power, learner performance or authority.

Fraud matrix/policy identities valid? (1=yes)	1.000000
Guarded modeled fraud policy cost reduction	63450.000000
Inference result identities valid? (1=yes)	1.000000
Guarded approximate two-sided p	0.882016
Planning integer/allocation identities valid? (1=yes)	1.000000
Guarded relative-target randomized users PER ARM	395482.000000

No real screening, user assignment, data collection, analytics/price/product changes, contact, data export, blocking, accusation, disclosure or spend. Missing original evidence, design and owner permission stop real action.

Evidence | Target prevalence and prospective design

Existing selected Penn State readings remain in course. This workbook derives the original synthetic D115/D116 cases at https://top1os.dev/; no new empirical source reading or measured deployment evidence.

Fraud input lineage	Fraud inputs -> Matrix -> Policy -> Audit. Prevalence must match target population; conditional rates may shift with fraud mix, threshold, season, selection or label quality. A case-control balanced sample is not a target base-rate estimate.
Inference input lineage	Experiment inputs -> Inference -> Audit. Counts from fictional independent users over one fixed horizon. Need actual eligibility, randomization, exposure, missing outcomes, interference and stopping records before any causal interpretation. No repeated-look validity inferred.
Planning input lineage	Planning inputs -> Planning -> Audit. Targets and z values are prospective stipulations. No power inferred from observed p or observed effect. Equal arms, normal approximation and noninformative attrition are not established real-world facts.
Strongest authored objections	Review may be imperfect and costly; fraud labels delayed or biased. Wald intervals can be poor near boundaries; count screen alone does not validate a design. p may round numerically to zero. Missing outcomes, clusters, multiple looks/variants or assignment bias need a different design/analysis.
Fictional day 0/day 30 hold gate	Validation/review/privacy and experiment/statistics/privacy roles need original labels/design, uncertainty, appeals and lawful use before a real decision. No actual person accepted roles. This is not a screening or experimentation grant.
Delayed changed-context task	Without notes change fraud prevalence/specificity/missed loss; rebuild matrix/odds/costs. Separately change group sizes/counts and prospective relative target/attrition; distinguish confidence interval from planned power and per arm from total. Record only actual errors, objections, correction effects and delayed transfer.

Actual work | No performed learner results

Actual attempts, confidence, source, errors, corrections, reviewer objection and delayed response remain blank.

Actual date / case	Attempt / confidence	Input / source	Observed error / reason	Correction / effect	Objection / reply	Delayed context / result

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Baseline matrix / Bayes / policy	PASS: independent baseline matrix/Bayes/cost/inference/planning checks.
Prevalence / conditional-rate shifts	PASS: prevalence 5%, specificity 99%, sensitivity 70%, restored.
Policy loss and review/friction costs	PASS: loss 200, review 40 and zero friction recomputed.
Boundary flags / zero-fraud states	PASS: no/all fraud, no/all flags; undefined ratios explicit.
Fixed-horizon counts / sample sizes	PASS: unequal arms and positive/negative/zero differences.
Low-count approximation holds	PASS: low/zero counts hold approximate interval/test summary.
Relative vs absolute planning targets	PASS: relative 4%, absolute 1 point, negative target and changed baseline.
Attrition / target zero / direction	PASS: attrition 0/20%, zero deltas make required n undefined.
Invalid inputs / case isolation	PASS: 17 invalid inputs block affected model; independent cases preserved.
Broken references / restored baseline	PASS: four broken refs blocked; baseline restored, actual learner blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.